

Estimating T-cell trafficking rates in glioblastoma: from clonal tracking to a continuous-time Markov chain

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We track T-cell clones across anatomical sites and time in glioblastoma and ask *at what rates cells move between sites and change phenotype*. We approach this in three layers, developed in order:

1. a **Bayesian phenotype-transition model** that estimates directed, per-tissue-pair transition probabilities with patient-level pooling (Sec. 2) — this is where “the rates we want” are made precise;
2. the **empirical transition matrix** P , a model-free estimate of the joint (site $\times$ phenotype) dynamics obtained directly from clonal tracking (Sec. 3);
3. the **continuous-time generator** Q with $P = \exp(Q \Delta t)$, which factors the dynamics into interpretable *migration*, *phenotype-transition*, and *expansion* rates (Sec. 4), with an open-system (birth/death) extension (Sec. 5).

This section fixes the notation used throughout.

1 Problem setup and definitions

1.1 From cells to tensors

Everything is built from one tagged object and a binning; each step adds a single definition.

1. **Cell.** Every T cell carries five tags: clone c (its TCR β), patient p , timepoint t , tissue s , phenotype k .
2. **Axes.** Cells are sorted on two categorical axes — tissue $S = \{\text{PBMC}, \text{CSF}, \text{TP}\}$ ($S = 3$) and phenotype ($K = 11$). Their product is the *joint state* $x = (s, k)$, with $N = SK = 33$ values, flattened site-major by $\iota(s, k) = sK + k$.
3. **Bin.** Fix a clone c and time t . The bin (s, k) holds one number: how many of c ’s cells are in tissue s with phenotype k .
4. **Count tensor.** Stacking the bins over time gives, *per clone*, a 3-D tensor of counts

$$n_c \in \mathbb{Z}_{\geq 0}^{T \times S \times K}, \quad n_c[t, s, k] = \#\{\text{cells of } c \text{ at time } t \text{ in bin } (s, k)\}.$$

Picture $T = 6$ time-slices, each an $S \times K = 3 \times 11$ grid of 33 bins (over all clones, a 4-D array $|\mathcal{C}| \times T \times S \times K$). Flattening $(s, k) \rightarrow x$ makes each slice a length- N vector $n_{c,t} \in \mathbb{Z}_{\geq 0}^N$.

5. **Clone total.** $N_{c,t} = \sum_x n_{c,t}(x)$ is how many cells of c were seen at t . Keep a slice only if $N_{c,t} \geq 2$.

6. **State vector.** Normalising a slice says *where the clone's mass sits*: $v_{c,t} = n_{c,t}/N_{c,t} \in \Delta^{N-1}$, a distribution over the 33 states (sums to 1).
7. **Observation.** A retained consecutive pair is one datum: $(v_{c,t}, v_{c,t+1})$ with weight $w_{c,t} = N_{c,t} + N_{c,t+1}$. Steps are one interval, $\Delta t = 1$ (timepoint index, not days), so all rates are **per interval**. The M pairs stack into $\mathbf{S}, \mathbf{D} \in \mathbb{R}^{M \times N}$ — the input to P (Sec. 3).

1.2 The Bayesian reduction

The Bayesian model (Sec. 2) looks at one *directed* tissue pair ($a \rightarrow b$), $a, b \in \mathcal{S}$, and one lineage ℓ (K_ℓ phenotypes: 7 for CD8, 4 for CD4; cross-lineage clones dropped). From a clone's count tensor it takes two phenotype vectors — the profile in the source tissue and the counts in the destination tissue one step later,

$$n_c^{\text{src}}(k) = n_c[t, a, k], \quad n_c^{\text{dst}}(k) = n_c[t+1, b, k] \in \mathbb{Z}_{\geq 0}^{K_\ell},$$

then forms the (smoothed, fixed) source distribution $\theta_c \propto n_c^{\text{src}} + 0.1 \in \Delta^{K_\ell-1}$ and the destination size $N_c = \sum_k n_c^{\text{dst}}(k)$. Learning how θ_c maps to n_c^{dst} folds migration and phenotype change into one matrix — which the CTMC later pulls apart.

1.3 Notation summary

Table 1 collects every symbol; objects defined in later sections (P , Q , and the three rate families) are listed with forward references.

1.4 The basic model at a glance

Figure 1 is the generative skeleton of the Bayesian model (Sec. 2), shown here to anchor the symbols above. Reading left-to-right is the generative flow: a fixed prior α and a latent pooling strength κ shape global transition rows T_k^g , which are sharpened into patient-specific rows $T_{p,k}^p$; each clone's observed source profile θ_c is pushed through its patient's matrix to predict its observed destination counts n_c^{dst} .

Variables in Figure 1.

α (fixed) Dirichlet prior concentration, $\alpha = K_\ell f^{\text{dst}}$ (floored at 0.5); centres the prior on the destination composition.

κ (latent) global→patient pooling strength, $\kappa \sim \text{Gamma}(5, 1)$; larger κ shrinks patients toward T^g .

T_k^g (latent) global transition row for source phenotype k , $T_k^g \sim \text{Dirichlet}(\alpha)$ — the population-level estimand.

T_k^p (latent) patient- p transition row, $T_k^p \sim \text{Dirichlet}(\kappa T_k^g)$.

θ_c (observed) clone source phenotype distribution in source tissue a .

n_c^{dst} (observed) clone destination counts in destination tissue b , $n_c^{\text{dst}} \sim \text{Multinomial}(N_c, \pi_c)$ with $\pi_c = \theta_c T_{p(c)}^p$.

N_c (fixed) destination cell total (Multinomial size).

The deterministic mean $\pi_c = \theta_c T_{p(c)}^p$ is folded into the two edges entering n_c^{dst} . *Figure generated by docs/figures/plate_basic.py.*

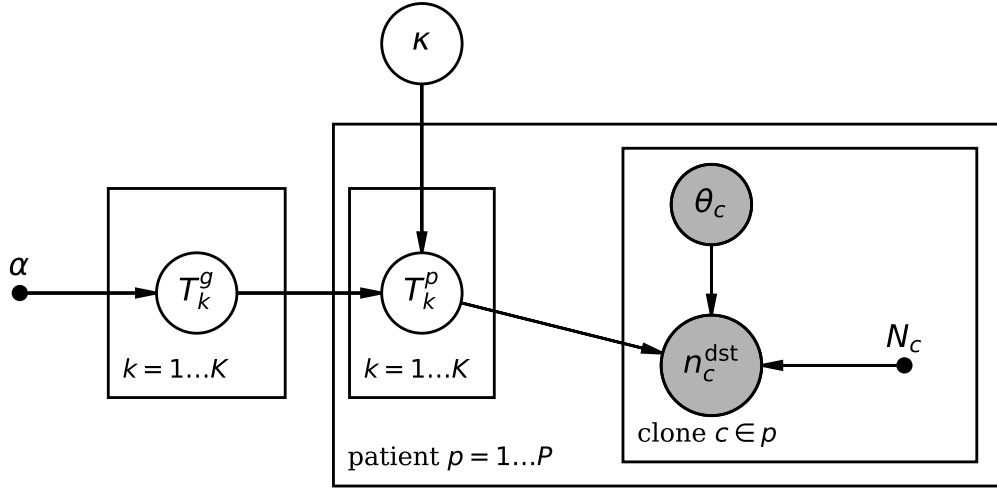


Figure 1: Plate diagram of the basic (per directed pair $a \rightarrow b$, per lineage ℓ) generative model. Open circles are latent (estimated); shaded circles are observed data, solid dots are fixed constants.

2 The Bayesian phenotype-transition model

[To be developed next: the full generative model, prior construction from f^{dst} , the hierarchical pooling via κ , the multinomial likelihood, and SVI inference.]

3 The empirical transition matrix P

[To be developed: clone-pair construction, the weighted least-squares estimator for P , and its interpretation.]

4 From P to the continuous-time generator Q

[To be developed: the closed CTMC, the migration/transition factorisation of Q , and the two estimators (matrix log and structured Frobenius fit).]

5 Open-system extension: birth and death

[To be developed: the augmented state with an unobserved compartment, and the exit/entry/expansion rates.]

Symbol	Meaning	Notes / value
<i>Indices and sets</i>		
$s, a, b \in \mathcal{S}$	tissue / site	$\{\text{PBMC, CSF, TP}\}$, $S = 3$
$k, u, v \in \mathcal{K}$	phenotype	$K = 11$ joint; K_ℓ per lineage
$\ell \in \{\text{CD8, CD4}\}$	lineage	switching clones removed
$x = (s, k) \in \mathcal{X}$	joint state	$N = SK = 33$
$\iota(s, k) = sK + k$	flat index (site-major)	$0, \dots, N - 1$
$c \in \mathcal{C}$	clone (TCR β)	lineage tracer
$p \in \{1, \dots, P\}$	patient	hierarchical grouping
$t \in \{1, \dots, T\}$	timepoint	$T = 6$; pairs $(t, t+1)$, $\Delta t = 1$
$(a \rightarrow b)$	directed tissue pair	$a, b \in \mathcal{S}$; Sec. 2
<i>Observed / fixed data</i>		
$n_{c,t}(s, k)$	cells of clone c at t in (s, k)	
$N_{c,t} = \sum n_{c,t}$	clone total at t	retained if ≥ 2
$v_{c,t} \in \Delta^{N-1}$	fractional state vector	$v = n_{c,t}/N_{c,t}$
$n_c^{\text{src}}, n_c^{\text{dst}} \in \mathbb{Z}_{\geq 0}^{K_\ell}$	clone phenotype counts in tissue a / b	
$\theta_c \in \Delta^{K_\ell-1}$	source phenotype distribution	$\theta_c \propto n_c^{\text{src}} + 0.1$
$N_c = \sum_v n_c^{\text{dst}}(v)$	destination cell total	Multinomial size
$f^{\text{dst}} \in \Delta^{K_\ell-1}$	destination phenotype composition in b	prior centre
<i>Parameters and estimands</i>		
$T_k^g, T_k \in \Delta^{K_\ell-1}$	global / pooled transition row	Sec. 2
$T_{p,k}^p \in \Delta^{K_\ell-1}$	patient- p transition row	Sec. 2
κ	global $\rightarrow$ patient pooling concentration	Gamma(5, 1)
$\alpha \in \mathbb{R}^{K_\ell}$	Dirichlet prior concentration	$\alpha = K_\ell f^{\text{dst}}$, floored at 0.5
$\pi_c = \theta_c T_{p(c)}^p$	expected destination distribution	deterministic
$P \in \mathbb{R}^{N \times N}$	empirical transition matrix	row-stochastic; Sec. 3
$Q \in \mathbb{R}^{N \times N}$	CTMC generator	$P = \exp(Q\Delta t)$; Sec. 4
$m_{a \rightarrow b}^{(k)}$	migration rate (site $a \rightarrow b$, phenotype k fixed)	Sec. 4
$r_{u \rightarrow v}^{(s)}$	phenotype-transition rate ($u \rightarrow v$, site s fixed)	Sec. 4
$g_{s,k}$	expansion (growth + / death -) rate	Sec. 5

Table 1: Master notation. The Bayesian model (Sec. 2) operates on the phenotype-only, per-lineage space (K_ℓ); the empirical matrix and generator (Secs. 3–5) operate on the full joint space ($N = 33$).